

Targeted and Logical Discovery of Piperazic Acid-Bearing Natural Products Based on Genomic and Spectroscopic Signatures

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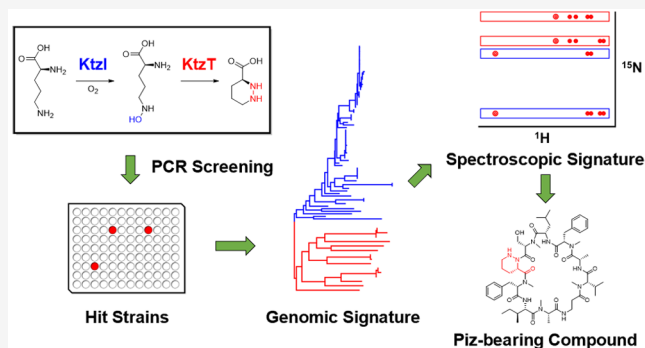


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ABSTRACT: A targeted and logical discovery method was devised for natural products containing piperazic acid (Piz), which is biosynthesized from ornithine by L-ornithine N-hydroxylase (KtzI) and N–N bond formation enzyme (KtzT). Genomic signature-based screening of a bacterial DNA library (2020 strains) using polymerase chain reaction (PCR) primers targeting *ktzT* identified 62 strains (3.1%). The PCR amplicons of *ktzT*-encoding genes were phylogenetically analyzed to classify the 23 clades into two monophyletic groups, I and II. Cultivating hit strains in media supplemented with $^{15}\text{NH}_4\text{Cl}$ and applying ^1H – ^{15}N heteronuclear multiple bond correlation (HMBC) along with ^1H – ^{15}N heteronuclear single quantum coherence (HSQC) and ^1H – ^{15}N HSQC-total correlation spectroscopy (HSQC-TOCSY) NMR experiments detected the spectroscopic signatures of Piz and modified Piz. Chemical investigation of the hit strains prioritized by genomic and spectroscopic signatures led to the identification of a new azinotrichin congener, polyoxyperuvin B seco acid (1), previously reported chloptosin (2) in group I, depsidomycin D (3) incorporating two dehydropiperazic acids (Dpz), and lenziamides A and B (4 and 5), structurally novel 31-membered cyclic decapeptides in group II. By consolidating the phylogenetic and chemical analyses, clade–structure relationships were elucidated for 19 of the 23 clades. Lenziamide A (4) inhibited STAT3 activation and induced G₂/M cell cycle arrest, apoptotic cell death, and tumor growth suppression in human colorectal cancer cells. Moreover, lenziamide A (4) resensitized 5-fluorouracil (5-FU) activity in both in vitro cell cultures and the in vivo 5-FU-resistant tumor xenograft mouse model. This work demonstrates that the genomic and spectroscopic signature-based searches provide an efficient and general strategy for new bioactive natural products containing specific structural motifs.



INTRODUCTION

Microbial secondary metabolites have significantly contributed to drug discovery and development by providing important clinical drugs representatively for the treatment of cancer, pathogenic infections, metabolic diseases, and immune system modulation.¹ Since the golden era of antibiotics, a random search for metabolites has been a prevailing strategy for discovering bioactive microbial compounds. The recent understanding of microbial compounds as genetically encoded small molecules has enabled heterologous expression or indigenous strain promoter engineering of biosynthetic gene clusters (BGCs), thus harvesting the metabolites produced by the targeted biosynthetic pathway.² However, these approaches require whole-genome sequences of microbial strains in the initial selection stage for biosynthetic pathways of interest, which limit their application to numerous microbial strains potentially harboring unknown BGCs. An alternative strategy is the polymerase chain reaction (PCR) screening of a microbial

DNA library with primer sets targeting specific biosynthetic genes for strain selection. New terpenoids,³ enediynes,⁴ and leinamycin derivatives⁵ were discovered based on strain prioritization by PCR primer screening of bacterial DNA libraries. Although this method efficiently applies to some microbial strains without complete genome sequences, it has two crucial limitations to overcome: (1) PCR screening only provides a selection of strains possessing the targeted genes. It does not guide the discovery of structurally novel compounds among these strains. (2) The detection of products biosynthe-

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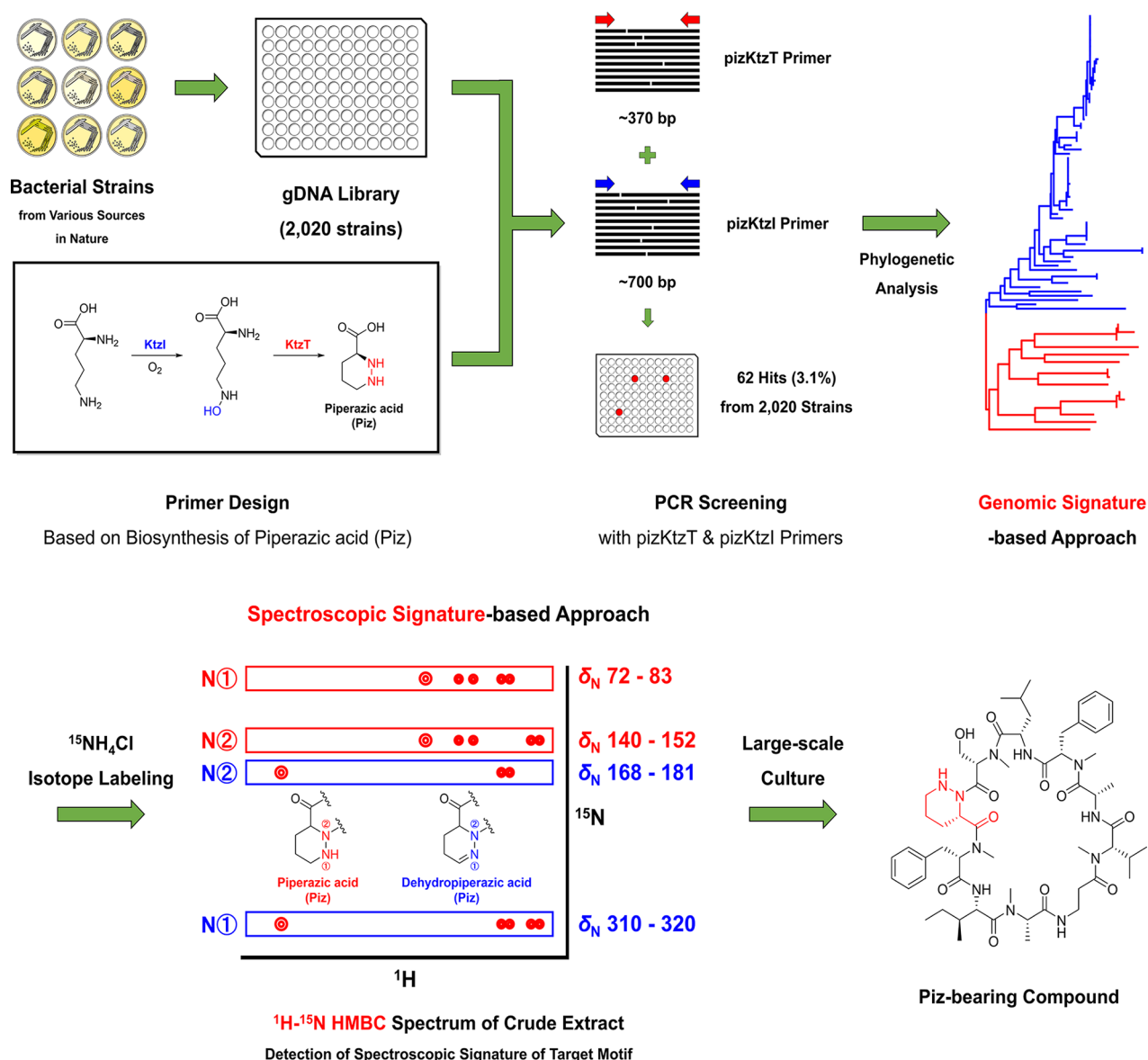


Figure 1. Overall process of genomic and spectroscopic signature-based discovery of structurally novel piperazic acid-bearing compounds.

sized by targeted genes is indirect or obscure because it currently relies on real-time PCR, an ambiguous search for metabolites with hypothetical molecular mass ranges, and a comparison of the chemical profiles of wild-type and PCR target gene knockout strains.

A direct NMR-based detection of desired biosynthetic products in crude culture extracts was reported with a new piperazic acid-bearing compound, incarnatapeptin, by feeding $^{15}\text{N}_2$ -L-ornithine as a precursor after identifying a biosynthetic gene cluster potentially producing a piperazic acid-bearing peptide in a strain among the NCBI databases of sequenced bacteria.⁶ In another example, whole-genome analysis coupled with a piperazate-targeted metabologenomic approach utilizing d^7 -L-ornithine and subsequent mass spectrometry-based detection led to the discovery of new dimeric piperazic acid-bearing biaryl-cyclohexapeptides, petrichorins A and B from the rare actinomycete *Lentzea flaviverrucosa* strain.⁷

In this context, a new strategy that (i) guides the discovery of structurally novel compounds possessing a target structural moiety from microbial strains without whole-genome data and

(ii) unequivocally detects the production of the desired compounds in microbial cultures at an early stage before purification in a more general manner for various structural motifs is required. Recently, in our targeted discovery of macrolactams, we utilized PCR amplicons (several hundred base pairs) of specific target genes, which provided a genomic signature to predict the types of macrolactams to be produced.⁸ We also developed a method to detect the spectroscopic signatures of macrolactams directly and unequivocally without purification and to differentiate the types of macrolactams produced by labeling metabolites with $^{15}\text{NH}_4\text{Cl}$ and subsequent ^1H - ^{15}N HSQC-total correlation spectroscopy (HSQC-TOCSY) NMR experiments. This study demonstrated that combining genomic and spectroscopic signatures can lead to a targeted discovery of macrolactam natural products.

Piperazic acid (Piz) is a unique hydrazine-containing amino acid rarely discovered in bacterial secondary metabolites.⁹ Piz is a cyclized amino acid and thus provides the structural rigidity of Piz-bearing compounds. The enhanced rigidity was considered as a factor increasing the biological activity because many of

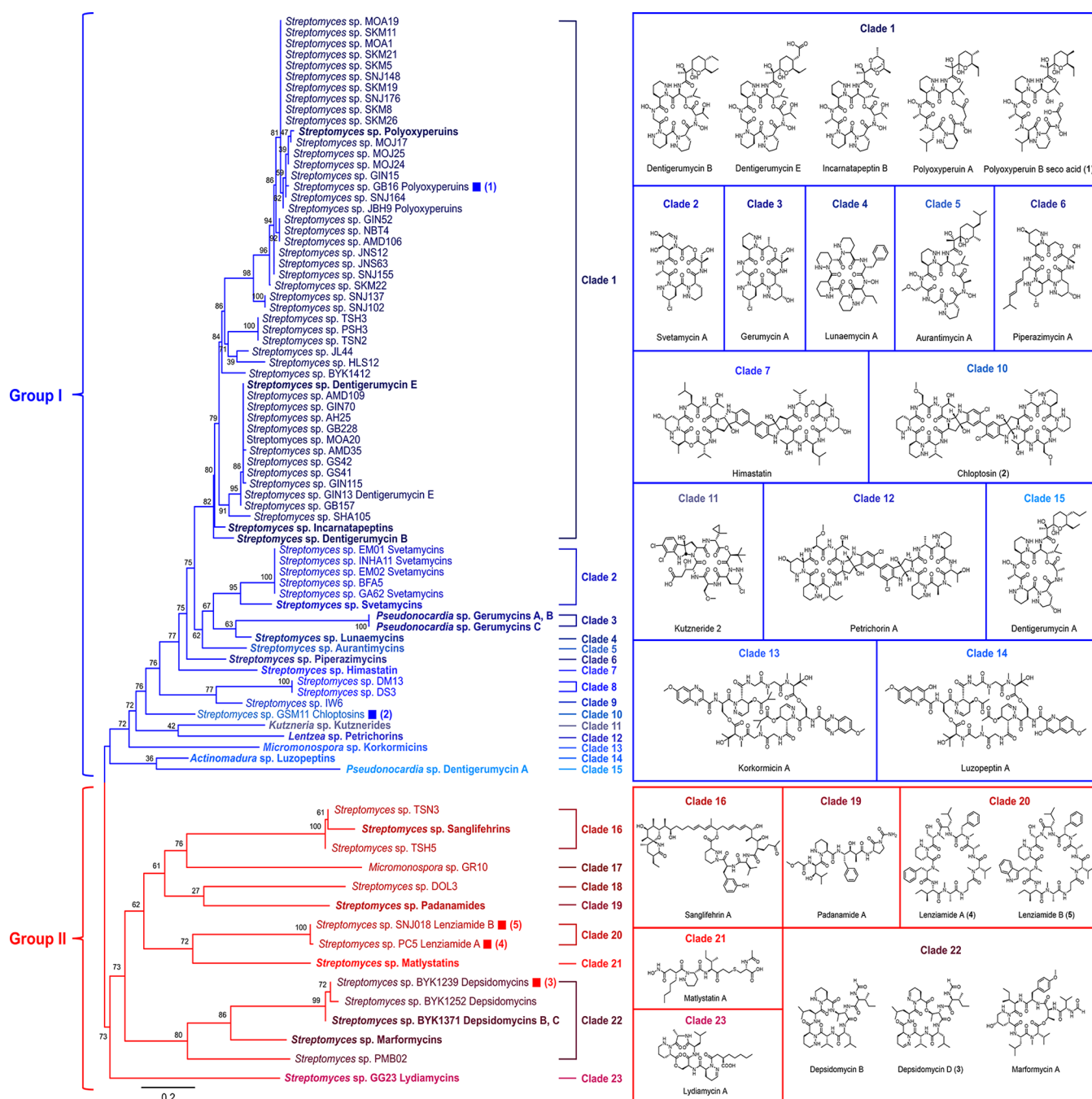


Figure 2. Phylogenetic tree of KtzT amplicons and structures of piperazic acid-bearing compounds. Shown in bold are the 22 reference strains that produce Piz-bearing compounds. Five strains and their Piz-bearing compounds investigated in this study are indicated with squares. Bootstrap supporting values are shown on the branches. Scale bar indicates the average number of substitutions per site.

these compounds exhibit potent anticancer and antibacterial activities.¹⁰ We also reported bioactive Piz-bearing compounds, antiproliferative dentigerumycin E,¹¹ antitumorigenic depsidomycins B–C,¹² and antitubercular lydiamycin A¹³ from *Streptomyces* spp. In our continuous search for Piz-containing compounds, we devised a targeted and logical discovery method for Piz-bearing bacterial metabolites. Because Piz is biosynthesized by the two key enzymes, L-ornithine N-hydroxylase (KtzI)¹⁴ and N–N bond formation enzyme (KtzT)¹⁵ and incorporated into parent molecules by specific adenylation domain,¹⁶ we utilized the genomic signatures of the two key genes, *ktzI* and *ktzT*, and the spectroscopic signature of various forms of Piz incorporated in bacterial metabolites detected by

¹H–¹⁵N two-dimensional (2D) NMR experiments to logically discover Piz-bearing compounds from microbial strains without whole-genome data.

Interestingly, the phylogenetic analysis of KtzT-encoding gene amplicons provided an atlas of Piz-bearing compounds in nature and guided the targeted and logical discovery of structurally novel bioactive compounds with Piz. ¹H–¹⁵N 2D NMR experiments coupled with ¹⁵NH₄Cl labeling were demonstrated to be a general method for identifying specific spectroscopic signatures of microbial metabolites possessing various nitrogenous structural moieties without chromatographic purification (Figure 1). By applying the genomic and

spectroscopic signature-based method, we discovered a new azinotrichin¹⁷ derivative, polyoxyperuvin B seco acid (1),¹⁸ a new depsidomycin¹² congener, depsidomycin D (3), and unprecedented 31-membered Piz-bearing cyclic decapeptides, lenziamides A and B (4 and 5) along with dereplication of chloptosin (2).¹⁹ Biological evaluation of the discovered compounds was performed using a cytotoxicity assay against human cancer cell lines, and their putative biosynthetic pathways were analyzed. Among the discovered compounds, lenziamide A (4) was further evaluated for its anticancer potential both in vitro and in vivo. Lenziamide A displayed selective antiproliferative activity against cancer cells compared with normal cells. The underlying molecular mechanism revealed that the antiproliferative activity of lenziamide A was related to the inhibition of STAT3 activation. STAT3 inhibition results in the induction of G₂/M phase cell cycle arrest and subsequently evokes apoptotic cell death in colon cancer cells. Lenziamide A also exhibited antitumor activity in vivo in animal models, suggesting the possibility of overcoming 5-FU drug resistance in 5-FU-resistant colon cancer cells.

■ RESULTS AND DISCUSSION

Genomic Signature-Based Screening of the Bacterial DNA Library. Two key enzymes form the Piz motif from the L-ornithine precursor: L-ornithine *N*-hydroxylase (KtZI)¹⁴ and *N*-*N* bond formation enzyme (KtZT).¹⁵ Genes for KtZI and KtZT are found in the biosynthetic gene clusters (BGCs) of Piz-bearing metabolites as genomic signatures (Figure S1). Because they may have evolved during the delineation of BGCs, we hypothesized that the sequences of KtZI- or/and KtZT-encoding genes could serve as a genomic signature to illustrate entire structural novelty and guide the targeted discovery of Piz-bearing metabolites without complete genome analysis.

To design the primers which target these genes in random strains without whole-genome data, the gene sequences of *ktzI* and *ktzT* homologues from the BGCs of 14 reported Piz-bearing compounds in the literature (aurantimycin,²⁰ dentigerumycin A,²¹ dentigerumycin B,²² dentigerumycin E,¹¹ gerumycin A,²³ gerumycin C,²³ himastatin,²⁴ kutzneride,²⁵ marformycin,²⁶ matlystatin,²⁷ padanamide,²⁸ sangliferhrin²⁹) and our own sequence data (lydiamycin¹³ and piperazimycin³⁰) were aligned and the conserved regions were identified (Figures S4 and S5). Two degenerate primers, pizKtZI and pizKtZT, were designed to target and amplify ~700 bp and ~370 bp conserved fragments within bacterial *ktzI* and *ktzT* gene homologues, respectively. In particular, we designed the degenerate primers specifically for KtZT-encoding genes because we had to avoid false hits by PaiB-family transcriptional regulators, which are not related to Piz biosynthesis but have significant amino acid similarity to the *N*-*N* bond formation enzyme (KtZT).³¹

The bacterial genomic DNA library comprised 48 strains from various environments, including intertidal mudflats, insect ecosystems, volcanic islands, Siberian ice wedges, saltern topsoil, and deep-sea sediments. The genomic DNA of 2020 strains was screened by PCR primarily using the pizKtZT primer set, and only the strains that showed a positive amplicon band from electrophoresis were cross-validated using the pizKtZI primer set (62 strains, 3.1%); the majority (98%) of the selected strains belonged to *Streptomyces* except for a rare actinomycete strain of *Micromonospora*. The genera *Actinomadura*, *Lentzea*, *Micromonospora*, *Pseudonocardia*, and *Kutzneria* were included in the phylogenetic tree as reference strains.

Phylogenetic Analysis of the *N*-*N* Bond Formation Enzyme (KtZT) and L-Ornithine *N*-Hydroxylase (KtZI) Gene Amplicons. The amplicons of the 62 strains generated using pizKtZT (sizes ranging from 363 to 375 bp) were subjected to phylogenetic analysis after translation into amino acid sequences. Including 22 references, these internal fragments in the KtZT homologues of the 84 strains could be divided into two major distinct groups and 23 clades when categorized by applying bootstrap support values of 80% as the threshold for clade designation (Figure 2). Interestingly, the phylogenetic analysis of KtZT homologues has provided an atlas of natural Piz-bearing compounds.

Clade 1 comprised 48 strains including the strains producing azinotrichin congeners such as dentigerumycins,^{11,22} incarnata-peptins,⁶ and polyoxyperuins.¹⁸ Clade 2 included a reference strain that produced svetamycins^{32,33} and five highly homologous strains. *Pseudonocardia* strains in clade 3 commonly produce gerumycins, which have a peptide core similar to azinotrichins but do not incorporate a polyketide-derived acyl chain.²³ Clade 4 is for the reference strain which biosynthesized lunaemycins, cyclic hexapeptides with four Piz residues.³⁴ Clade 5 represents a strain producing aurantimycin,²⁰ which also belongs to the azinotrichin family. The reference strain, reported to produce 18-membered cyclic hexapeptides piperazimycins,³⁰ was assigned to clade 6. The producer of himastatin, an aryl–aryl coupled hexapeptide dimer, was classified into clade 7.²⁴ Two highly similar *Streptomyces* strains with unknown metabolites were assigned to clade 8. *Streptomyces* sp. IW6, isolated from the Siberian ice wedge, is the only member of clade 9. *Streptomyces* GSM11 is the only member of clade 10. Clades 11–15 were singleton clades with each reference strain. The kutzneride-producing reference strain was assigned to clade 11.²⁵ The *Lentzea* strain which produced dimeric piperazyl cyclopeptides, petrichorins, is in clade 12.⁷ Clades 13 and 14 contain the producers of diester-type dimeric bisintercalator, korkormicins and luzopeptins, respectively.³⁵ *Pseudonocardia* sp. in clade 15 produced dentigerumycin A in the azinotrichin family.²¹ One of the strains in clade 16 was the reference strain producing sangliferhrins,²⁹ which are mainly polyketide-driven compounds bearing Piz. *Micromonospora* sp. GR10, isolated from the blister beetle, was assigned as the only member of clade 17. Clade 18 contained only one strain, and its metabolites have not been investigated. The padanamide-producing strain was assigned to clade 19.²⁸ Clade 20 comprised two strains: *Streptomyces* sp. PC5 was isolated from the dung beetle *Onthophagus lenzii*, and SNJ018 was collected from marine sediment. The strain in clade 21 produced matlystatin.²⁷ The strains producing 19-membered aldehyde-bearing depsipeptides, depsidomycin¹² and marformycin²⁶ were classified in clade 22. Finally, lydiamycin-producing strain, *Streptomyces* sp. GG23,¹³ is the only member of clade 23.

These clades could be phylogenetically grouped into two monophyletic groups: clades 1–15 were assigned to group I, whereas clades 16–23 were categorized into group II. Group I contained a large clade of strains producing azinotrichin congeners (dentigerumycins, polyoxyperuins, and incarnata-peptins), which form the largest family of Piz-bearing compounds. The compounds in this family share the common structural backbone of a 19-membered hexadepsipeptide with an polyketide-derived acyl side chain. Approximately half of the strains, including the reference strains, were classified into clade 1, suggesting that the strains that possess BGCs of azinotrichin family compounds are dominant among all Piz-BGC-bearing

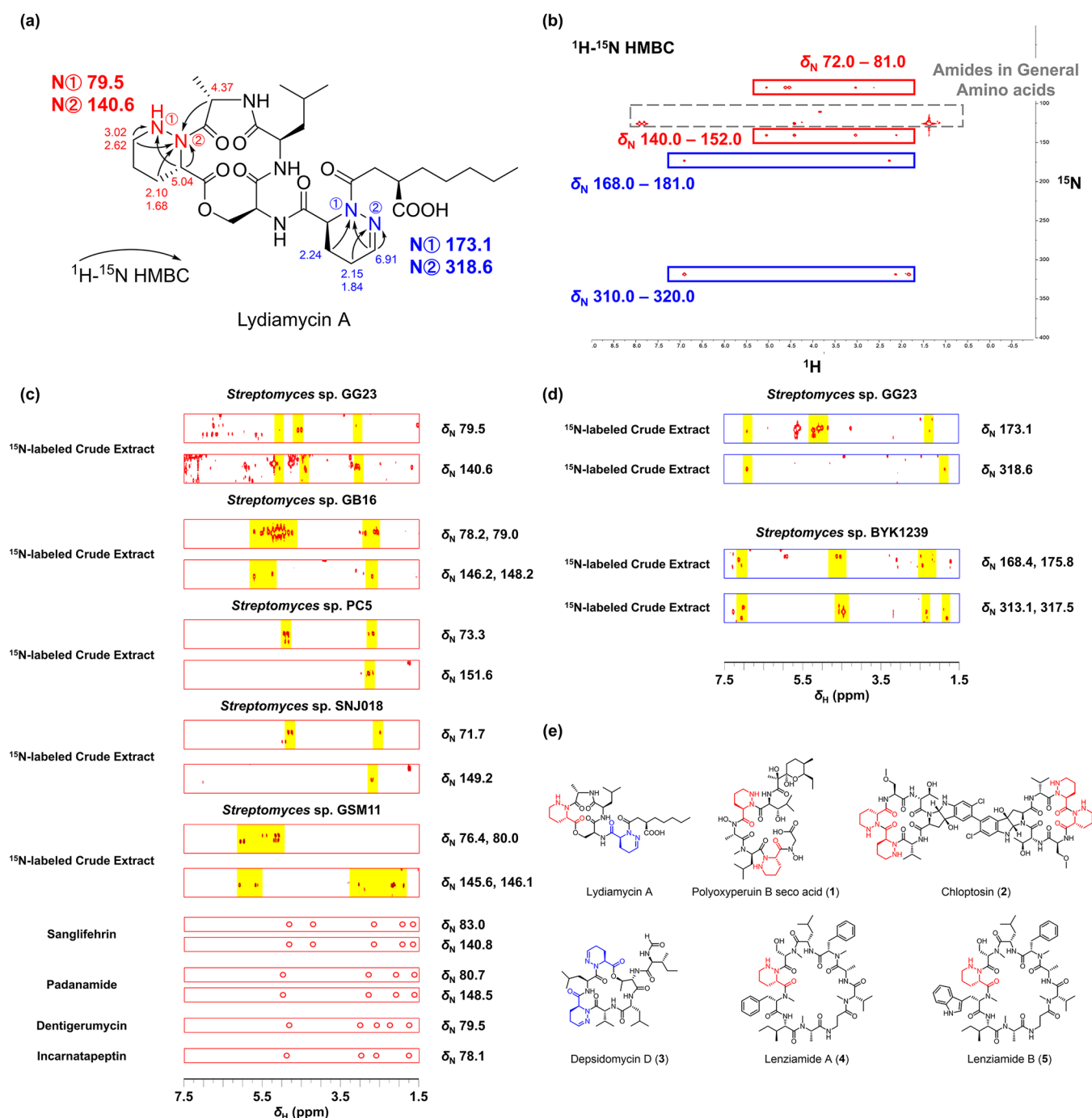


Figure 3. (a) ¹H and ¹⁵N chemical shifts of lydiamycin A. (b) ¹H-¹⁵N HMBC spectrum of lydiamycin A. (c) Expanded ¹H-¹⁵N HMBC spectrum around δ_N of nitrogens in piperazine acid. (d) Expanded ¹H-¹⁵N HMBC spectrum around δ_N of nitrogens in dehydropiperazine acid. (e) Structures of the piperazine acid-containing compounds from hit strains.

strains in nature. This may also explain why most Piz-bearing compounds discovered were azinothricin family compounds. Therefore, the structures of the group I compounds can be largely defined as the azinothricin congeners (clades 1, 5, and 15) containing a 19-membered hexadepsipeptide and a polyketide-derived acyl chain, the 18-membered cyclic hexapeptide-based compounds (clades 2, 3, 4, 6, 7, 10, 11, and 12), and the diester-type dimeric peptides (clades 13 and 14). Although there were much fewer strains (15 strains) than group I (69 strains), group II had greater structural diversity for Piz-bearing metabolites. Group II has 8 clades (clades 16–23) with two structurally unidentified clades (clades 17 and 18). Group II are

structurally diverse and distinguished from group I by not containing any structural types belonging to group I. These results indicate that the phylogenetic analysis of ~370 bp KtzT-encoding gene amplicons could prioritize hit strains and thus guide the discovery of structurally novel Piz-bearing compounds.

Phylogenetic analysis of KtzI-encoding gene amplicons also provided an analogous result. However, some of the references did not fall into the appropriate group or clade (Figure S8), probably because KtzI is unessential and conditionally dispensable for Piz production.³¹ This result indicated that

KtzT provides a better genomic signature for predicting the structures of Piz-bearing natural products.

Spectroscopic Signature-Based Screening of Hit Strains. We cultivated 69 hit strains selected by genomic signature-based screening and obtained their culture time-course chemical profiles using LC/MS. Analysis of the profiles showed that 12 strains (GB16, JBH9, and GIN13 in clade 1; EM01, INHA11, EM02, and GA62 in clade 2; GSM11 in clade 10; PC5 and SNJ018 in clade 20; and BYK1239 and BYK1252 in clade 22) produced peptide-like metabolites (19% of 62 strains) based on ultraviolet (UV) and MS spectra. Rapid dereplication using our in-house UV library, liquid chromatography–mass spectrometry (LC/MS) operating software, and literature data identified known compounds. GIN13 produced dentigerumycin E, previously reported from the reference strain JB5.¹¹ JBH9's chemical profile was identical to that of GB16. BYK1252 is chemically identical to BYK1239 in the same clade. EM01, INHA11, EM02, and GA62 commonly produce svetamycins, 18-membered halogenated cyclic depsipeptides bearing three piperazic acid-derived amino acid residues.^{32,33}

This result was consistent with the clade–structure relationship based on the atlas of Piz-bearing compounds generated by the KtzT-encoding gene amplicons (Figure 2). Initial dereplication enabled us to exclude the seven strains, GIN13, JBH9, EM01, INHA11, EM02, GA62, and BYK1252, from further chemical studies and focused on five strains, GB16 in clade 1, GSM11 in clade 10, PC5 and SNJ018 in clade 20, and BYK1239 in clade 22.

Although the five selected strains (GB16, GSM11, BYK1239, PC5, and SNJ018) produced peptide-like metabolites by LC/MS analysis, it was not unequivocally confirmed that these metabolites were Piz-bearing compounds. To clarify if Piz-bearing metabolites are produced, we initially attempted to utilize ¹⁵N₂-L-ornithine feeding and ¹H–¹⁵N HSQC-TOCSY experiments, as previously reported in the discovery of incarnatapeptins by Ryan and Andersen's groups.⁶ However, this method has an apparent limit in detecting modified Piz. Piz can be modified into approximately 20 derivatives.⁹ Piz-derived amino acids retaining the “–N–NH–” partial structures are detectable by ¹H–¹⁵N HSQC-TOCSY experiments, but some derivatives, such as dehydropiperazic acid (Dpz), are deficient of NH because the “–N–NH–” moiety can be converted to “–N–N=” by a multitasking cytochrome P450 (Luz26 found in the luzopeptin biosynthesis),³⁵ which cannot be detected by ¹H–¹⁵N HSQC-TOCSY NMR experiments, limiting the detecting method. Therefore, the spectroscopic signature of Piz should be expanded for Piz-derived motifs with either “–N–NH–” or “–N–N=” and a more general detecting method is in demand.

To develop a spectroscopic signature for Piz-derived motifs with either “–N–NH–” or “–N–N=”, analysis of ¹⁵N chemical shifts of Piz-bearing compounds is required. However, ¹⁵N chemical shifts have rarely been reported in Piz-bearing compounds, with only sanglifehrins,³⁶ padanamides,³⁷ dentigerumycins,³⁸ and incarnatapeptins⁶ being reported. Moreover, the ¹⁵N chemical shifts of the nitrogen atoms in the dehydropiperazic acids (Dpz) have never been reported before. Therefore, we fed the lydiamycin A-producing *Streptomyces* sp. GG23 with ¹⁵N₂-L-ornithine because lydiamycin A contains regular Piz and Dpz. The ¹H–¹⁵N HSQC, ¹H–¹⁵N HSQC-TOCSY, and ¹H–¹⁵N heteronuclear multiple bond correlation (HMBC) NMR spectra were acquired for the GG23 extract. The four

nitrogen atoms displayed distinct ¹⁵N chemical shifts at δ_N [79.5 (NH) and 140.6 (amide N) in Piz] and [173.1 (amide N) and 318.6 (–N=) in Dpz]. Analysis of the ¹⁵N chemical shifts of Pizs in sanglifehrins, padanamides, dentigerumycins, incarnatapeptins, and lydiamycins enabled the extraction of the spectroscopic signature of Piz-derived units, mainly by ¹H–¹⁵N HMBC NMR spectra, as shown in Figure 3, suggesting that ¹H–¹⁵N HMBC NMR experiments are beneficial for detecting spectroscopic signatures, along with ¹H–¹⁵N HSQC and ¹H–¹⁵N HSQC-TOCSY NMR spectra.

Five selected strains (GB16, GSM11, BYK1239, PC5, and SNJ018) were cultured in media supplemented with ¹⁵N₂-L-ornithine. The EtOAc extracts of the strains were subjected to ¹H–¹⁵N HSQC and ¹H–¹⁵N HSQC-TOCSY NMR analyses, and the spectroscopic signature derived from the NH in the Piz motif was monitored. The crude extract of two strains, GSM11 and BYK1239, showed the spectroscopic signature of NH in the region of δ_N 72.0–83.0 and δ_H 3.80–5.40 in the ¹H–¹⁵N HSQC spectra. In addition, protons in the Piz motif's spin systems were observed in the region of δ_N 72.0–83.0 and δ_H 0.30–5.50 in the ¹H–¹⁵N HSQC-TOCSY spectra. However, even though the mass spectra showed isotopically labeled features, no signal was observed in the ¹H–¹⁵N HSQC and HSQC-TOCSY NMR experiments for GB16 and SNJ018. Only weak NH 1-bond signals were detected in the ¹H–¹⁵N HSQC spectrum of PC5, possibly due to poor labeling efficiency, low production yield, unwanted signal broadening of NH, and the insensitivity of ¹H–¹⁵N NMR experiments, indicating the limitations of this method utilizing the ¹⁵N-enriched expensive precursor, ¹⁵N₂-L-ornithine.

To develop a clearer, economical, and general spectroscopic signature-based screening method for Piz-bearing metabolites and any nitrogenous natural products, we utilized ¹⁵NH₄Cl, which is significantly more economical than ¹⁵N₂-L-ornithine (see the Supporting Information). The five selected strains were then cultivated in ¹⁵NH₄Cl-enriched media, and their EtOAc extracts were subjected to ¹H–¹⁵N HMBC along with ¹H–¹⁵N HSQC and ¹H–¹⁵N HSQC-TOCSY NMR experiments (Figures 3 and S30–S51). In the 2D ¹H–¹⁵N NMR spectra, N–N bond nitrogen atoms resonated in the region of δ_N 72.0–83.0, while amide nitrogens of the N–N bond were detected at δ_N 140.0–152.0, of which range is close to but different from the chemical shifts (δ_N 105.0–133.0) of ordinary secondary amide nitrogen atoms. ¹H–¹⁵N HMBC NMR experiments have a crucial advantage in detecting the imine nitrogen when Piz is converted to Dpz, which is not observable by ¹H–¹⁵N HSQC or ¹H–¹⁵N HSQC-TOCSY experiments. The spectroscopic signature of an imine in the Dpz motif was characteristic of the region of δ_N 310.0–320.0 in the ¹H–¹⁵N HMBC spectrum of BYK1239. When Dpz was formed, the spectroscopic signature of the amide nitrogen atoms in Piz was detected at δ_N 168.0–181.0. Therefore, the five strains (GB16, GSM11, BYK1239, PC5, and SNJ018) were confirmed to produce Piz-bearing compounds.

Predicting and Guiding the Discovery of Structurally New Piz-Bearing Compounds by Phylogenetic Analysis.

Through large-scale cultivation of the selected five strains following the spectroscopic signature, chromatographic isolation, and spectroscopic analysis, mainly based on NMR experiments, we elucidated the structures of the Piz-bearing compounds. GB16 belongs to clade 1, which includes azinothricin family compounds. Therefore, GB16 was predicted

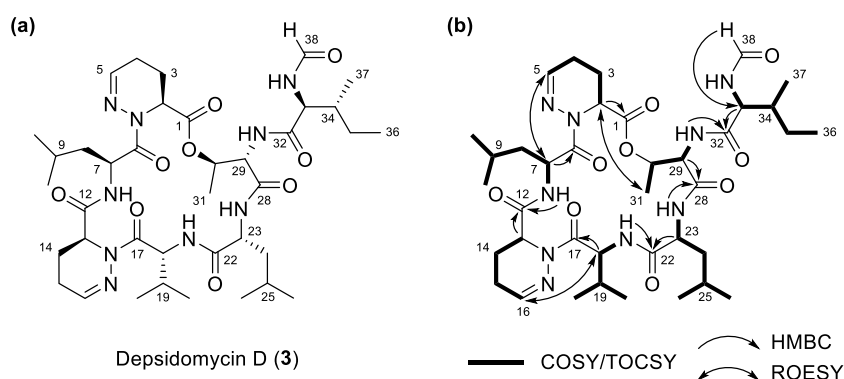


Figure 4. (a) Structure of depsidomycin D (3). (b) Key COSY/TOCSY, HMBC, and ROESY correlations of 3.

to biosynthesize an azinotrichin congener. The planar structure of the major compound from GB16, polyoxyperuvin B seco acid (1), was identical to that of a recently reported azinotrichin congener, polyoxyperuvin A seco acid, bearing a polyketide-derived pyran and hexapeptide chain with two piperazic acid units.¹⁸ The absolute configuration of polyoxyperuvin B seco acid (1) was established using advanced Marfey's analysis,³⁹ *J*-based configuration analysis,⁴⁰ and BGC analysis utilizing antiSMASH 6.0.⁴¹ Polyoxyperuvin B seco acid (1) is stereochemically different from polyoxyperuvin A seco acid and is thus a new compound bearing *N*-Me-D-Leu and *S*-Piz instead of *N*-Me-L-Leu and *R*-Piz.¹⁸ Cultivation of GSM11 in clade 10 yielded the previously reported chloptosin (2), a dimeric cyclohexapeptide, based on NMR and MS spectroscopic analyses.¹⁹

BYK1239, which is most closely related to the depsidomycin-producing *Streptomyces* sp. BYK1371¹² in clade 22, enabled us to predict that this strain produces compounds similar to depsidomycin. Interestingly, the BYK1239 extract showed a signature of dehydropiperazic acid (Dpz) at δ_N 313–318. As a Dpz-bearing compound has not been reported in the depsidomycin family, we focused on the Dpz signature and identified a new compound, depsidomycin D (3). Because depsidomycin D (3) showed very similar NMR spectra to those of depsidomycin B which we reported from *Streptomyces* sp. BYK1371 in the same clade (Figure 2), we compared the ¹H and ¹³C NMR data of the two compounds (Figure S116 and Table S8). A detailed analysis of the 2D NMR data revealed that depsidomycin D (3) shares five identical amino acid units with depsidomycin B: *N*-formylisoleucine (fIle), threonine (Thr), two leucines (Leu-1 and Leu-2), and valine (Val). In contrast, two double-bond protons (H-5 (δ_H 7.09) and H-16 (δ_H 7.06)) were observed in the ¹H NMR data of 3 instead of the aliphatic δ -protons in the regular Piz units. In addition, two double-bonded carbons, C-5 (δ_C 145.5) and C-16 (δ_C 144.7), exhibited HSQC correlations with H-5 and H-16, respectively. Correlation spectroscopy (COSY) and TOCSY data showed that H-2 to H-5 belong to one spin system, and H-13 to H-16 belong to a single spin system. In addition, these protons showed ¹H–¹⁵N HMBC correlations with nitrogens at δ_N 168.4, 175.8, 313.1, and 317.5, which were the spectroscopic signatures of Dpz. As predicted, these one-dimensional (1D) and 2D NMR data showed that depsidomycin D (3) contained Dpz units instead of Piz units. ROESY correlations of α -amino protons in Dpz showed that the Dpz-1 unit was located between the Thr and Leu-1 units and the Dpz-2 unit was assigned between the Leu-1 and Val units. Consequently, the structure of depsidomycin D

(3) was determined as a new cyclic depsipeptide with an *N*-formyl group and two Dpz units (Figure 4).

To date, a compound bearing two Dpz units has been rarely reported. Depsidomycin D (3) was a rare example incorporating two dehydropiperazic acid residues. Depsidomycin D was configurationally determined using the advanced Marfey's method,³⁹ followed by liquid chromatography/mass spectrometry (LC/MS) analysis.

Lastly, strains PC5 and SNJ018, which form clade 20, are closely related; however, there is no chemical information about these two strains. Because they are closely related in the KtzT-encoding gene amplicon phylogenetic tree (Figure 2), they were predicted to produce similar Piz-bearing compounds, even though their origins are very different (PC5 was isolated from a dung beetle, whereas SNJ018 was collected from marine sediment). Detailed chemical analyses of PC5 and SNJ018 revealed that their major metabolite was actinomycin D, an approved anticancer drug. However, large-scale cultivation of these strains led to the discovery of a new cyclic peptide bearing the Piz residue, lenziamide A (4). SNJ018 yielded a new compound, lenziamide B (5).

Lenziamide A (4) was obtained as a white amorphous solid, and its formula was assigned as C₅₇H₈₇N₁₁O₁₁ (20° of unsaturation) by HRMS and ¹H and ¹³C NMR spectroscopic data. The ¹H and HSQC NMR data indicated that 4 has 6 exchangeable protons (δ_H 8.79, 7.73, 7.22, 6.73, 4.91, and 4.77), 10 aromatic protons (δ_H 7.31–6.41), 15 *N*/*O*-bound methine/methylene protons (δ_H 6.00–2.68), 15 *N*-methyl protons (δ_H 3.43, 3.33, 2.96, 2.75, and 2.74), 3 aliphatic methine protons (δ_H 2.37, 2.23, and 1.57), 14 aliphatic methylene protons (δ_H 3.58–0.06), and 24 aliphatic methyl protons (δ_H 1.39–0.68).

The ¹³C NMR and HSQC data showed that 4 has 10 carbonyl carbons (δ_C 176.2–169.6), 12 aromatic carbons (δ_C 139.9–127.3), 12 *N*/*O*-bound methine/methylene carbons (δ_C 69.9–41.4), 5 *N*-methyl carbons (δ_C 39.0, 32.4, 31.8, 30.0, and 28.6), 3 aliphatic methine carbons (δ_C 35.4, 25.1, and 24.8), 7 aliphatic methylene carbons (δ_C 37.2–19.4), and 8 aliphatic methyl carbons (δ_C 24.2–9.6). All 1-bond C–H correlations were assigned using ¹H, ¹³C, and HSQC NMR spectroscopic analyses.

Comprehensive analyses of the COSY, TOCSY, and HMBC NMR data indicated that 4 possesses 10 amino acid units. HMBC correlations between α -protons and carbonyl carbons revealed that 4 contained residues of two alanine (Ala), two phenylalanines (Phe), leucine (Leu), serine (Ser), isoleucine (Ile), β -alanine (β -Ala), and valine (Val). Protons H-25, H-26a, H-26b, H-27a, H-27b, H-28a, H-28b, and 28-NH were found to belong to a single spin system through analyses of the COSY,

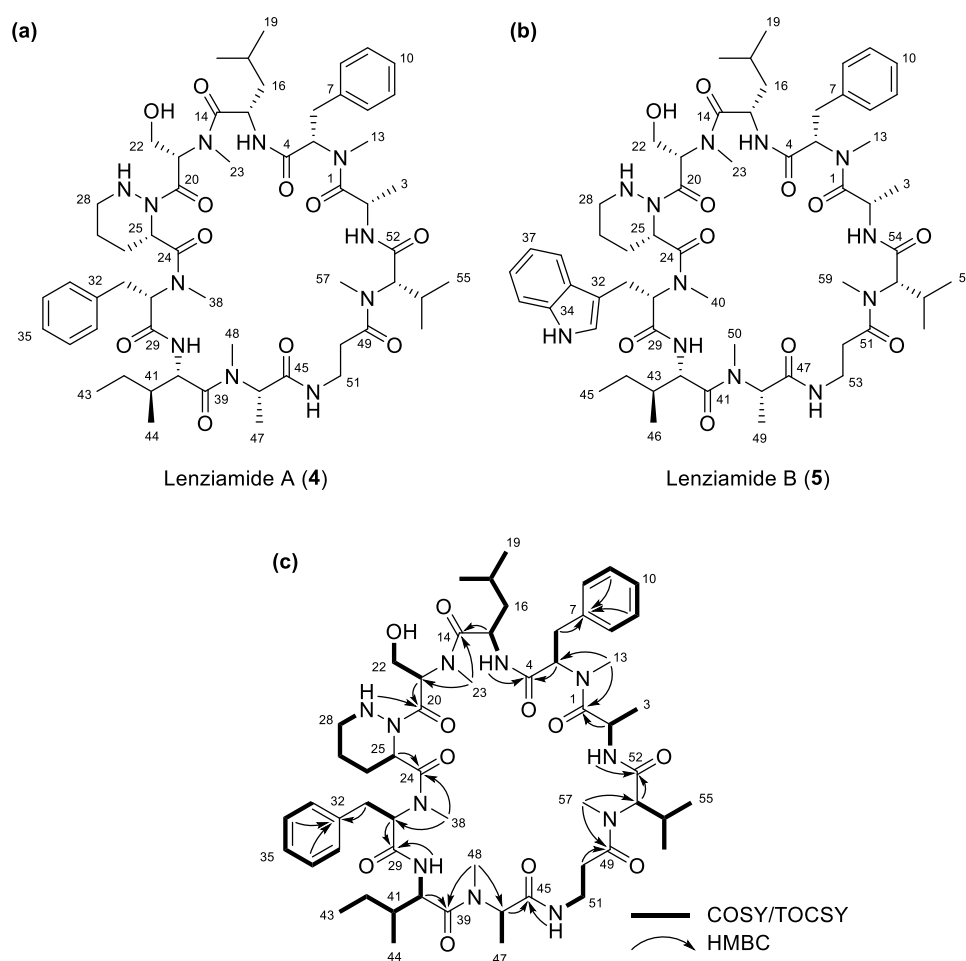


Figure 5. (a) Structure of lenziamide A (4). (b) Structure of lenziamide B (5). (c) Key COSY/TOCSY and HMBC correlations of lenziamide A.

TOCSY, and ^1H – ^{15}N HSQC–TOCSY data. Also, these protons showed ^1H – ^{15}N HMBC correlations with the nitrogen atoms at δ_{N} 73.3 and 151.6, which were the spectroscopic signatures of Piz. Based on ^{15}N NMR data, we identified that 4 contained a single Piz unit. HMBC correlations from *N*-methyl protons (H_3 -13, H_3 -23, H_3 -38, H_3 -48, and H_3 -57) to the corresponding α -carbons identified the positions of the *N*-methyl groups to construct *N*-Me-Phe-1, *N*-Me-Ser, *N*-Me-Phe-2, *N*-Me-Ala, and *N*-Me-Val. Further analysis of HMBC correlations assembled the 10 amino acids to elucidate the structure of 4. H_3 -13 (δ_{H} 3.33) showed HMBC correlations with C-1 (δ_{C} 176.2) and C-5 (δ_{C} 59.0), which connected Ala and *N*-Me-Phe-1. HMBC correlations from H-5 and 15-NH to C-4 secured the connectivity from *N*-Me-Phe-1 to Leu. This peptide chain was extended to *N*-Me-Ser by H_3 -23 (δ_{H} 2.96)/C-14 (δ_{C} 176.2) and H_3 -23/C-21 (δ_{C} 69.9) heteronuclear couplings. Piz was then assembled to *N*-Me-Ser based on the HMBC correlations from H-21 and 28-NH to C-20. HMBC correlations from H_3 -38 (δ_{H} 2.75) to C-24 (δ_{C} 172.3) and C-30 (δ_{C} 63.3) linked Piz and *N*-Me-Phe-2. The *N*-Me-Phe-2-Ile connectivity was supported by the HMBC correlations from H-30 and 40-NH to C-29. Ile and *N*-Me-Ala were coupled by 3-bond correlations from H_3 -48 (δ_{H} 3.43) to C-39 (δ_{C} 172.3) and C-46 (δ_{C} 52.7). 51-NH/C-45 and H-46/C-45 HMBC correlations connected β -Ala next to *N*-Me-Ala. The last *N*-methyl protons, H_3 -57 (δ_{H} 2.74), showed HMBC correlations to C-49 (δ_{C} 173.6) and C-53 (δ_{C} 62.4), allowing for the connectivity from β -Ala to *N*-Me-Val. Finally, *N*-Me-Val was linked to Ala based on 2-NH/C-52 HMBC

correlation, thus completing the cyclic decapeptide structure of 4 depicted in Figure 5 (Ala-*N*-Me-Phe-1-Leu-*N*-Me-Ser-Piz-*N*-Me-Phe-2-Ile-*N*-Me-Ala- β -Ala-*N*-Me-Val).

Lenziamide B (5) differed from lenziamide A in that it possesses *N*-Me-Trp instead of *N*-Me-Phe in the cycle. The absolute configurations of lenziamides A and B were assigned using the advanced Marfey's method³⁹ and GITC (2,3,4,6-tetra-*O*-acetyl- β -D-glucopyranosyl isothiocyanate) derivatization,⁴² followed by LC/MS analysis. Lenziamides A and B (4 and 5) are structurally novel and possess the largest cyclic structure among Piz-bearing natural products, a 31-membered ring with 10 amino acids, including nonproteinogenic β -alanine, along with five *N*-methyl amino acids. These compounds do not belong to any previously reported structural class, as expected from the KtzT-encoding gene amplicon sequences, demonstrating that phylogenetic analysis of the ~370 bp KtzT-encoding gene amplicons can guide the discovery of novel Piz-bearing compounds.

Analysis of Putative Biosynthetic Gene Clusters Producing Piz-Bearing Compounds. Whole-genome sequencing of the depsidomycin-producing *Streptomyces* sp. BYK1239 was performed. Based on the whole-genome analysis of strain BYK1239 using antiSMASH 6.0,⁴¹ a putative nonribosomal peptide synthetase (NRPS) gene cluster of depsidomycins was identified. The BGC of depsidomycin shows 54% similarity to that of marformycin, which contains a piperazine acid and formylated moieties.²⁶ The BGC of depsidomycin includes three clustered NRPS genes (*ddmE*,

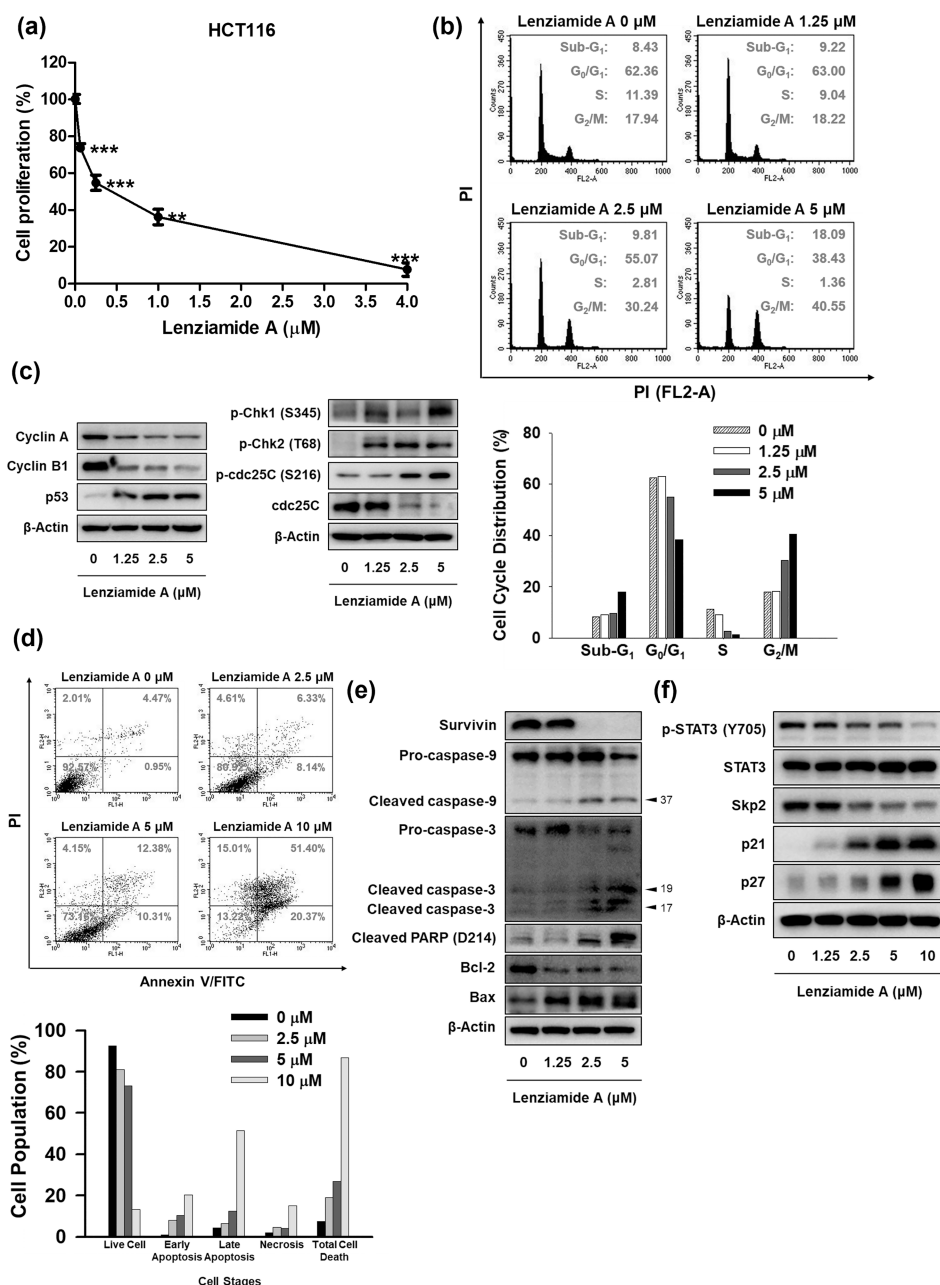


Figure 6. In vitro efficacy of lenziamide A (4). (a) HCT116 cells were treated with lenziamide A (4) for 72 h, and the cell proliferation was measured using the SRB assay. (b) HCT116 cells were treated with lenziamide A (4) at the concentrations specified for 24 h. Following staining with PI for 30 min, the distribution of cells in each cell cycle phase was determined using flow cytometry. (c) HCT116 cells were treated with the indicated concentrations of lenziamide A (4) for 24 h and lysed, and the expression levels of indicated proteins were determined using western blotting analysis. β -Actin was used as an internal control. (d) HCT116 cells were treated with the indicated concentrations of lenziamide A (4) for 48 h and stained with annexin V-FITC and PI. The stained cells were subsequently analyzed through flow cytometry. (e) HCT116 cells were treated with the indicated concentrations of lenziamide A (4) for 48 h and lysed, and the levels of indicated proteins were determined using western blotting analysis. β -Actin was used as an internal control. (f) HCT116 cells were treated with the indicated concentrations of lenziamide A (4) for 24 h and lysed, and the levels of indicated proteins were determined using western blotting analysis. β -Actin was used as an internal control.

ddmF, and *ddmG*), two Piz biosynthesis-related genes (*ddmJ* and *ddmK*), and a gene (*ddmB*) for *N*-formyl groups (Figure S121). The NRPSs (DdmE, DdmF, and DdmG) comprised 7 modules corresponding to the incorporation of 7 amino acid units into depsidomycin D. Each Ddm module contains a minimal set of NRPS domains which is composed of condensation (C), adenylation (A), and thiolation (T) domains, and two additional epimerization (E) and a thioesterase (TE) domain are in modules 3, 4, and 7, respectively. The predicted substrate

specificities of A domains are all consistent with the modular structure of depsidomycin D. Modules 3 and 4 contains E domains, and C domains of modules 4 and 5 are ^DC_L type. These are consistent with the determined absolute configuration of depsidomycin D (3) by the advanced Marfey's analysis. Depsidomycins B and C were also found along with 3 in *Streptomyces* sp. BYK1239. However, the absolute configurations of depsidomycins B and C, which were previously reported from *Streptomyces* sp. BYK1371, were inconsistent with that of 3

and the genetic analysis, requiring a further study for structural revision particularly in their configurations. Then, we performed advanced Marfey's analysis for depsidomycin B and revised the absolute configurations of Leu, Thr, and Ile (Figure S116). For the formation of the formyl group, there is no putative formylation domain within the first module. Instead, the methionyl-tRNA formyltransferase homologue DdmB was found in the gene cluster. The DdmB showed high sequence identity (80%) to the methionyl-tRNA formyltransferase (MfnA) from marformycin BGC. Therefore, DdmB is expected to work with the starter C domain of module 1 to catalyze the *N*-formylation on the first amino acid L-Ile, in a similar manner to marformycin biosynthesis.²⁶ DdmJ and DdmK, which individually show 51 and 50% sequence identities with KtzI (encoding L-ornithine *N*-hydroxylase) and KtzT (*N*-*N* bond formation enzyme) from the kutzneride biosynthetic pathway,^{14,15} are expected to form the dehydropiperazic acid (Dpz) incorporated by modules 5 and 7. Lastly, the TE domain in module 7 catalyzes macrocyclization between Dpz and L-Thr units originated from modules 7 and 2, respectively, yielding depsidomycin D. A gene encoding a Luz26 homologue, which was reported as multi-tasking cytochrome P450 to convert Piz to Dpz in luzopeptin,³⁵ was not identified in the BGC for 3, indicating that the production of Dpz may have an alternative pathway.

We also analyzed the whole-genome sequences of lenziamides A and B (4 and 5)-producing *Streptomyces* spp. PC5 and SNJ018. Utilization of antiSMASH 6.0⁴¹ enabled the identification of putative BGCs (each *lnz*p and *lnz*s, together *lnz*) of 4 and 5. Although the *lnz*s BGC from SNJ018 has a few more genes, the two *lnz* BGCs are almost identical, specifically including a single NRPS gene (*lnzD*), piperazic acid biosynthetic genes (*lnzC* and *lnzB*), and a β -alanine formation gene (*lnzA*) (Figure S122). The putative function of *lnzA* is an aspartate 1-decarboxylase, which catalyzes the decarboxylation of aspartate to produce β -alanine. *lnzC* and *lnzB* share 49 and 50% sequence identities with KtzI and KtzT,^{14,15} respectively. The NRPS *lnzD* is composed of 10 modules that correspond to each amino acid unit in structures 4 and 5. Each module contains C-A-T domain motifs, and additional *N*-methyltransferase (MT) domains are found in modules 2, 4, 6, 8, and 10, which transfer methyl groups to the *N*-terminus of amino acids. Even though the predictions of amino acid activation specificity of all A domains are not consistent with the actual lenziamide structure, A domains of *lnz* NRPS machinery activate 10 amino acids in the following order: L-Ala, L-Phe, L-Leu, L-Ser, L-Piz, L-Phe, L-Ile, L-Ala, β -Ala, and L-Val. In particular, the SNJ018 strain produces additional lenziamide B along with lenziamide A, unlike the PC5 strain. While the amino acid-specific codes of the other A domains from the both *lnz*Ds are all the same, the amino acid-specific code of A6 domain of *lnz*sD (DGWSVASVDK) is distinct from that of A6 domain of *lnz*pD (DGWSVAAYDK). Therefore, it is expected that the *lnz*sD A6 domain, which has a more flexible substrate specificity, activates not only Phe but also Trp, enabling the production of Trp-containing lenziamide B as well. Finally, a TE domain located at the C-terminus of module 10 catalyzes macrocyclization between L-Ala and L-Val unit originated from modules 1 and 10, respectively, resulting in the production of lenziamides (Figure S122).

Antiproliferative Activities of Lenziamide A in Human Cancer Cells. To determine the biological activity of these novel natural Piz-bearing compounds, the antiproliferative activities of compounds 1–5 were evaluated against a panel of human cancer cell lines. Compounds 2, 4, and 5 exhibited

antiproliferative activity against human cancer cells in vitro (Table S19). Chloptosin (2) showed the most potent growth-inhibitory activity against cancer cells; however, the growth of normal human cell lines (MRC-5 and CCD 841 CoN) was also strongly inhibited by chloptosin with similar IC₅₀ values. However, the antiproliferative activities of lenziamide A (4) and lenziamide B (5) were shown to be relatively selective against cancer cells compared to normal cells, with selective index (SI) values (IC₅₀ value in cancer cells/IC₅₀ value in normal cells) ranging from 23 to 89. Since compound 4 exhibited a higher SI value and a relatively larger amount was isolated than compound 5, molecular mechanistic studies were performed using lenziamide A (4) with quality control (99% purity). Based on the most potent growth-inhibitory activity of lenziamide A against HCT116 colorectal cancer cells, its antiproliferative activities against additional colon cancer cell lines were evaluated. Lenziamide A effectively suppressed the growth of all tested colorectal cancer cells (Table S20). To further elucidate the antiproliferative activity of lenziamide A in human colorectal cancer cells, HCT116 cells, which showed the highest sensitivity to lenziamide A, were selected for a detailed molecular mechanistic study. Lenziamide A exhibited concentration-dependent antiproliferative activity against HCT116 cells (Figure 6a).

Cell cycle analysis revealed that treatment with lenziamide A for 24 h induced the accumulation of cells in the G₂/M phase in a concentration-dependent manner, suggesting that the antiproliferative activity of lenziamide A is, in part, associated with the induction of G₂/M phase arrest in colon cancer cells (Figure 6b). Further studies were conducted to determine the expression of proteins related to G₂/M phase cell cycle regulation by lenziamide A. Since HCT116 cells possess wild-type p53 status,⁴³ G₂/M arrest is mainly associated with decreased cyclin A and B1 expression.^{44,45} Lenziamide A effectively downregulated cyclin A and cyclin B1 expression and augmented p53 levels. In addition, cdc25C, a crucial mediator of mitosis,⁴⁶ was inactivated (phosphorylated) by the upregulation of p-chk1 and p-chk2 following lenziamide A treatment (Figure 6c). Further analysis investigated whether lenziamide A induced apoptotic cell death in HCT116 cells. In response to a 48 h treatment with lenziamide A, a remarkable accumulation of cells in the sub-G₁ phase was detected (Figure S123).

Further analysis investigated whether lenziamide A induced apoptotic cell death in HCT116 cells. HCT116 cells were treated with lenziamide A for 48 h and stained with annexin V-FITC and PI. As shown in Figure 6d, the early apoptotic (annexin V+/PI-) cell populations were increased to 8.14, 10.31, and 20.37% with 2.5, 5, and 10 μ M lenziamide A treatment, respectively. The late apoptotic (annexin V+/PI+) cell populations were also increased to 6.33, 12.38, and 51.40% with 2.5, 5, and 10 μ M lenziamide A treatment, respectively. The induction of apoptosis by lenziamide A was also confirmed by the regulation of apoptosis-associated protein expression, with modulation of caspase and PARP cleavage and survivin, Bcl-2, and Bax expression (Figure 6e). The activated-STAT3 signaling is associated with cell cycle regulation and apoptotic cell death in human cancer cells.⁴⁷ Therefore, we evaluated whether lenziamide A affected activated-STAT3 (phospho-STAT3) and its downstream targets. We found that the expression of phosphorylated STAT3 (activated form) and its downstream target Skp2 was effectively downregulated. In contrast, the

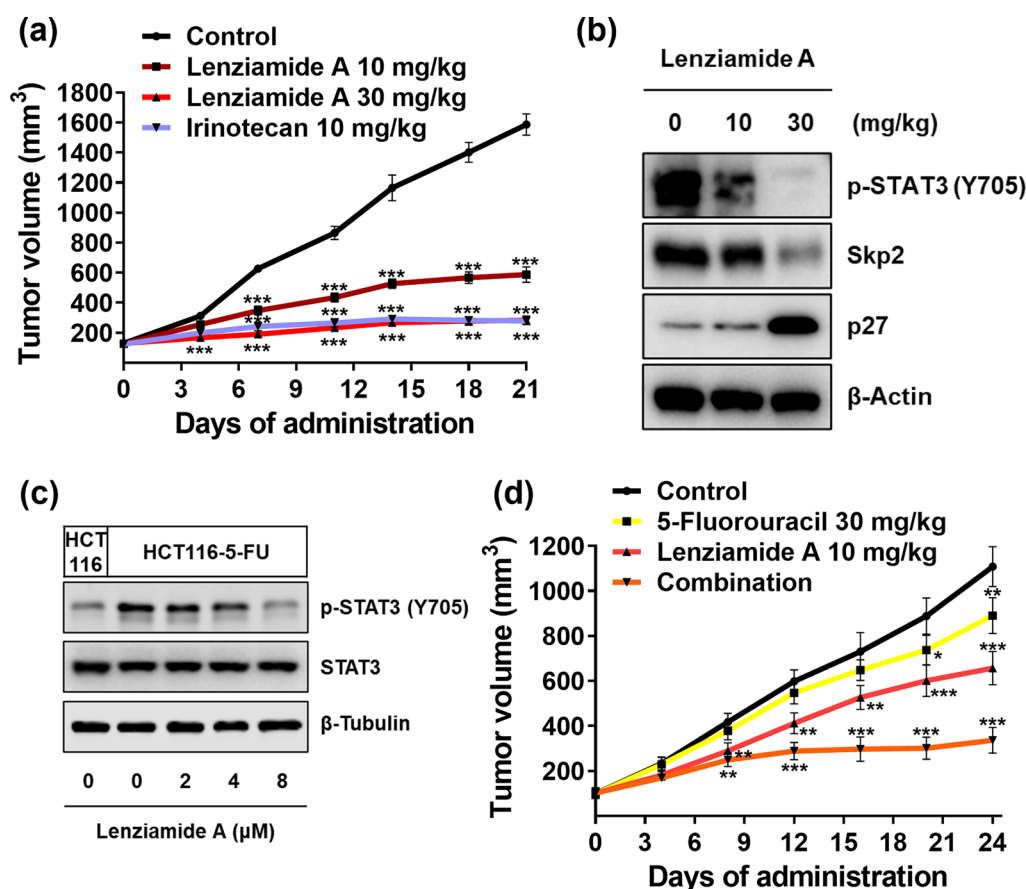


Figure 7. In vivo efficacy of lenziamide A (4). (a) Tumor volumes of the HCT116 xenograft mouse model, which were administered intraperitoneally three times per week with the vehicle, lenziamide A (4) (10 or 30 mg/kg), and irinotecan (10 mg/kg) for 21 days, measured every 3–4 days with a digital slide caliper. Irinotecan was used as a positive control. (b) Small portions of tumors from each group were homogenized in complete lysis buffer (active motif). The expression levels of indicated proteins were determined using western blotting analysis. β -Actin was used as an internal control. (c) The cells were treated with the indicated concentrations of lenziamide A (4) for 24 h and lysed and the levels of indicated proteins were determined using western blotting analysis. β -Tubulin was used as an internal control. (d) Tumor volumes of the HCT116-5-FU xenograft mouse model, which were administered intraperitoneally three times per week with the vehicle, lenziamide A (4) (10 mg/kg), 5-fluorouracil (30 mg/kg), and a combination of both compounds for 24 days, measured every 4 days with a digital slide caliper.

expression of the CDK inhibitors p21 and p27⁴⁸ was upregulated following lenziamide A treatment (Figure 6f).

Antitumor Efficacy of Lenziamide A In Vivo. We determined the in vivo antitumor efficacy of lenziamide A in a xenograft nude mouse model implanted with HCT116 cells. Lenziamide A exhibited effective antitumor activity in colon cancers, with a tumor growth inhibition ratio of 63.03% and 82.21% on the termination day in the 10 and 30 mg/kg lenziamide A-administered groups, respectively, compared to the vehicle-treated control group (Figure 7a). No significant changes in body weight or over toxicity were observed in the lenziamide A-treated groups (Figure S124a). Consistent with the in vitro results (Figure 6f), biochemical analysis of the tumor tissues also revealed that the expression of p-STAT3 (Y705) and Skp2 was downregulated, whereas that of the CDK inhibitor p27 was upregulated by lenziamide A treatment (Figure 7b), suggesting that lenziamide A effectively modulates the STAT3-Skp2 axis in the tumor growth.

Overcoming 5-FU Resistance with Lenziamide A. The standard regimen for advanced and metastatic colorectal cancers (CRCs) is generally employed in clinic by 5-fluorouracil (5-FU)-based combination therapies such as FOLFOX (leucovorin + 5-FU + oxaliplatin) and FOLFIRI (leucovorin + 5-FU + irinotecan).⁴⁹ However, aggressive and advanced types of CRCs

often exhibit resistance to 5-FU, which is a major cause of chemotherapy failure.⁵⁰ To evaluate whether lenziamide A can overcome the resistance of 5-FU in colon cancer cells, a 5-FU-resistant HCT116 cell line (HCT116-5-FU) was established by continuously exposing parental drug-sensitive HCT116 cells to 5-FU. HCT116-5-FU cells exhibited approximately 100-fold resistance to 5-FU compared to the parental HCT116 cells (Table S21). Multidrug resistance (MDR) is a phenotype of cancer cells with reduced sensitivity to diverse structurally unrelated drugs.⁵¹ Interestingly, HCT116-5-FU cells showed acquired resistance to structurally unrelated drugs used to treat CRC, including irinotecan and etoposide. However, lenziamide A exerted potent antiproliferative activity with similar IC₅₀ values in both cell lines. These data suggest that lenziamide A can overcome 5-FU resistance in HCT116-5-FU cells. We further evaluated whether lenziamide A could resensitize resistant cells to 5-FU. Cotreatment with lenziamide A and 5-FU resulted in a higher antiproliferative activity than any other treatment alone. The synergistic effects were calculated by combination index (CI) analysis using the Chou–Talalay method⁵² (Table S22).

Recently, aberrant activation of STAT3 has been considered to be correlated with acquired drug resistance.^{53,54} Interestingly, HCT116-5-FU cells also exhibited hyperactivation of STAT3

compared to their parent HCT116 cells. However, lenziamide A treatment for 24 h effectively suppressed p-STAT3 expression in HCT116-5-FU cells, indicating the possibility of overcoming acquired anticancer drug resistance in CRCs (Figure 7c). We evaluated the in vivo antitumor efficacy of lenziamide A in a xenograft model implanted with HCT116-5-FU cells. Consistently with the in vitro data, combined treatment with 5-FU and lenziamide A notably inhibited tumor growth in vivo compared to treatment with each compound without toxicity (Figures 7d and S124b).

CONCLUSIONS

The genomic signature of the key gene encoding the *N*–*N* bond formation (KtzT) enzyme responsible for the biosynthesis of piperazic acids enabled screening of our bacterial DNA library using PCR primers and efficiently selecting hit strains that could potentially produce Piz-bearing compounds. This PCR-based method may have limitations that the PCR screening can be performed within the ranges of collected library strains, and not every hit strain produces desired Piz-bearing compounds. However, it can be easily applicable for environmental bacterial strains without whole-genome data. In addition, it can be more efficient to produce and obtain reasonable amounts of targeted compounds than metagenomic approaches, followed by heterologous expression, which requires complicated genomic works. Detailed phylogenetic analysis of the KtzT-encoding gene PCR amplicon sequences allowed 23 clades to be classified comprising two monophyletic groups, predicted the structures of Piz-bearing compounds, and thus guided the logical chemical investigation of novel Piz-bearing compounds. For the unequivocal detection of compounds incorporating various modified forms of piperazic acids, such as the dehydropiperazic acid (Dpz), without purification, we adopted isotopic labeling with $^{15}\text{NH}_4\text{Cl}$ and utilized ^1H – ^{15}N HMBC along with ^1H – ^{15}N HSQC-TOCSY and ^1H – ^{15}N HSQC NMR experiments, which provided a diagnostic spectroscopic signature at δ_{N} 72–83 for Piz and δ_{N} 310–320 for Dpz.

Initial dereplication and detailed chemical analysis based on the spectroscopic signature of selected hit strains in group I identified polyoxypiperin seco acid (1) in clade 1, svetamycin in clade 2, chloptosin (2) in clade 10, and previously reported Piz-bearing compounds. Group II contained fewer strains, but they were chemically diverse. Further chemical examination of the strain belonging to clade 22, which contained a depsidomycin-producing strain, resulted in the discovery of depsidomycin D (3), a rare Piz-bearing compound with two dehydropiperazic acids (Dpzs). The distinct phylogenetic clade 20 contains two closely related *Streptomyces* strains, PC5 and SNJ018, which produce structurally novel 31-membered cyclic decapeptides, lenziamides A and B (4 and 5). Lenziamides constitute the largest cycle among Piz-bearing compounds. In addition, the putative nonribosomal peptide synthetase (NRPS) biosynthetic pathways were identified by bioinformatic analysis of the whole genome of the new compound producers.

More importantly, by consolidating the phylogenetic and chemical analyses, we elucidated 19 clade–structure relationships among the 23 clades. Therefore, a phylogenetic and structural atlas of Piz-bearing compounds was constructed to facilitate the targeted and logical discovery of Piz-bearing compounds from bacterial strains without whole-genome data.

To explore the biological activity of the isolated Piz-bearing compounds, the novel lenziamide A (4) has shown therapeutic potential in terms of anticancer activity. In particular, lenziamide

A exhibited potent growth-inhibitory activity against human colon cancer cells and showed high selectivity compared with normal colon cells. Additionally, lenziamide A significantly inhibited the growth of 5-FU resistant colon cancer cells, both in vitro and in vivo, which may be beneficial for overcoming drug resistance in clinical settings. A molecular mechanism study revealed that the anticancer activity of lenziamide A is partly associated with the inhibition of STAT3 activation, which induces cell cycle arrest and apoptosis in colon cancer cells. These findings suggest that the novel Piz-bearing lenziamide A may be prioritized as a lead compound in developing anticancer agents.

In addition, we applied ^{15}N labeling with $^{15}\text{NH}_4\text{Cl}$ and ^1H – ^{15}N HMBC NMR experiments to detect nitro group-bearing compounds. We successfully identified the signature at δ_{N} 350–400 of lajollamycin^{55,56} possessing a nitro group⁵⁷ in its extract without chromatographic purification (Figure S72). Therefore, using ^1H – ^{15}N HMBC NMR experiments coupled with ^{15}N enrichment by supplementing $^{15}\text{NH}_4\text{Cl}$ in culture media can be a general method to detect the spectroscopic signatures of structural motifs without protonated nitrogen. Our successful targeted and logical discovery of Piz-bearing compounds, along with our previous works targeting macrolactams⁸ and enediynes,⁵⁸ suggests that a genomic and spectroscopic signature-based search for a particular motif could be an efficient and general research strategy in natural product discovery.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.3c04699>.

Experimental procedures, compound characterizations, 1D and 2D NMR spectra, chromatographic data, genomic analysis data, and computational studies (PDF)

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Notes

The authors declare the following competing financial interest(s): The authors submitted patent applications for lenziamides and their anticancer activity (KR10-2389725, PCT/KR2020/012966, and US20220402970A1).

[‡]Deceased on December 19, 2020.

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